Simulink Laboratory Exercise 01: Signal Processing & Electrical System Modeling

Learning Objectives

1. Analyze the differences between discrete-time signal generators and continuous sources in Simulink, and configure data exchange with the MATLAB Workspace.
2. Formulate differential equations (ODEs) for dynamic systems (RL circuit), convert them into a block-oriented simulation structure, and evaluate system behavior in the time domain.
3. Build physical models using Simscape and critically validate the results against classic signal-based modeling.

Task 1.1: Signal Sources, Workspace Coupling & Data Processing

Context & Problem Statement: For an automated test bench control system, measurement signals must be generated, stored in memory, and subsequently analyzed in MATLAB. Which signal generator is suitable for which application, and how can data be passed to post-processing without data loss?

Phase 1: Preparation & Pre-Lab

Research the **Sine Wave** and **Signal Generator** blocks in the MATLAB documentation.

Question: How does mathematical signal generation differ between these two blocks (e.g., regarding sample time and computational efficiency)?

Phase 2: Hands-On Modeling (In-Lab)

Step A: Model Setup & Signal Visualization

Create a new Simulink model named **Exercise_1_1_model.slx**.
Place a **Sine Wave** block and a **Signal Generator** block (configured to Sine) into the model with the following parameters:

• Amplitude: 10 V • Frequency: 10 Hz (T = 100 ms) • Simulation Time: 250 ms

Visualize both signals stacked vertically in a split **Scope**.

Experiment: Clear the MATLAB Workspace and run the simulation again.
Analysis: Inspect the vector 'tout' in the Command Window. What does this vector contain, and how is its step size determined?

Step B: Signal Routing & Workspace Export

Combine both signals using a **Mux** block and connect the output to a single-input Scope.
Name the signal lines before the Mux block by double-clicking on them (e.g., **u_sine** and **u_siggen**). Verify whether the signal names appear in the Scope legend.
Export the multiplexed signal using the **To Workspace** block:

- Variable Name: **Sinewave**
- Save format: Test both **Timeseries** and **Array**.

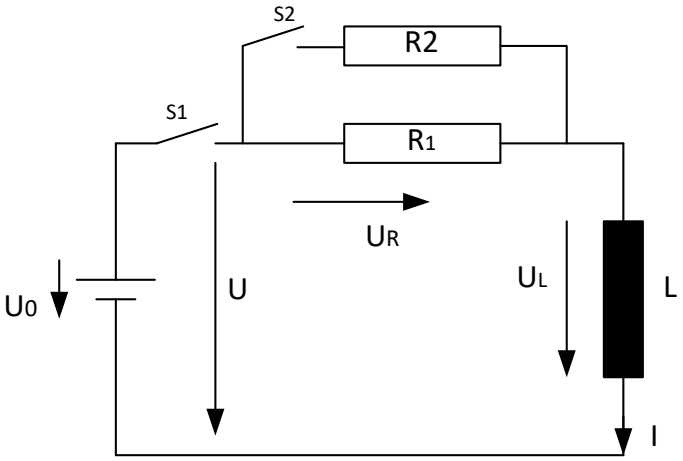
Phase 3: Post-Processing & Reflection (Post-Lab)

Create a MATLAB Live Script named **Exercise_1_1.mlx**.
Write a script to plot the matrix **Sinewave** against the time vector **tout** in a customized figure (including axis labels and legend).

Question: When choosing the 'Array' save format – why is time information lost in the export? Which alternative blocks (e.g., **Clock**) or data structures are needed to store time and signal values together in a single matrix?

Task 1.2: Physical System Modeling of an RL Circuit

Context & Problem Statement: An electromagnetic actuator is modeled as an RL circuit (R = 10 Ω, L = 1 mH, U_0 = 10 V). Transient current responses occur during switching events and must be evaluated through both mathematical and physical simulation.



Phase 1: Mathematical Modeling (Pre-Lab)

Formulate the mesh equation for the RL circuit and solve for the highest derivative of current (di/dt):

Text

Create an initialization script **Exercise_1_2_init.m** to define parameters R, L, and U_0 in the MATLAB Workspace.

Phase 2: Block-Oriented Simulation in Simulink

Step A: Base Model

Implement the ODE (Ordinary Differential Equation) in Simulink (**Exercise_1_2_model.slx**) using Integrator, Gain, and Sum blocks.
Simulate a step voltage input (U_0 = 10 V switches on at t = 100 ms; stop time: 300 ms).
Display the current response i(t) on a Scope.

Step B: Parameter Study (Resistance Variation)

Vary the resistance R systematically from 2 Ω to 10 Ω in steps of 2 Ω.

How does the system time constant τ = L / R visually change on the Scope?

Step C: Extension – Dynamic Switching Event

Extended Scenario: At t = 150 ms, a second resistor R_2 = 100 Ω is switched into the circuit (switch S2 closes).

Extend your Simulink model to incorporate this switch (e.g., using a **Switch** block) and analyze the transient current drop or rise.

Phase 3: Physical Modeling with Simscape & Validation

Build a new model **Exercise_1_2_model_sc.slx** using the **Simscape / Electrical** library (Resistor, Inductor, DC Voltage Source, Ideal Switch, Electrical Reference, Solver Configuration).

Measure the current using a **Current Sensor** and send the signal to a Scope via a **PS-Simulink Converter**.

Validation & Reflection:

- Compare the results from the Simulink ODE model with the Simscape physical simulation.

- Which modeling approach is less error-prone for complex networks, and why?